Design Space Exploration for the Design of Reliable SRAM-based FPGA Systems

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Abstract

This paper presents a design space exploration approach for the design of SRAM-based FPGAs with Single Event Upsets fault mitigation capabilities, exploiting the feature of partial, dynamic reconfiguration offered by the target platform. In fact, in this scenario, several techniques are available and they can be applied at different granularity levels; thus, we propose a support for exploring the different alternatives by means of a framework taking into account several figures of merit, based on genetic algorithms. The several elements taken into account for evaluating the different solutions are here analyzed and discussed, and the application of the methodology and framework to some case studies is reported, allowing both a tuning of the metrics and the analysis of the identified most interesting solutions, which do not correspond, in general, to the trivial ones.

1 Introduction

In the last decade, several studies have been proposed to mitigate the effects of Single Event Upset (SEU) faults in SRAM-based FPGAs, proposing periodic scrubbing of configuration bitstreams [1], triplication and voting [2, 3], information redundancy via Error Detecting And Correcting (EDAC) codes [4, 5], or duplication with comparison [6], when erroneous data can be discarded and recomputed.

These reliability-oriented techniques based on fault detection and/or correction can be efficiently coupled with partial dynamic reconfiguration [7] to achieve SEU mitigation properties [6, 8]; yet, the task is not an easy one. In fact, when designing a reliable system, the designer usually selects one of the available techniques and applies it to either the entire system (using a coarse grain granularity), or to every module in it (using a fine grain granularity); in a few cases only, a solution in between is adopted, and is designed ad hoc, based on the designer's knowledge of the various modules. In general, to the authors' knowledge, there is no available tool for exploring the solution space corresponding to intermediate granularity levels, where sets of modules are clustered together and the reliability oriented technique is applied to the entire cluster as if it were a unique component.

Thus, given the combination of available techniques and the possible granularity levels they can be applied at, the designer faces too many solutions, that can hardly be compared manually; our previous work [6, 8] was based on the generation of a large number of promising solutions, to be then analyzed with respect to a few parameters, requiring a

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significant time in terms of effort. Furthermore, changing the metrics taken into account to investigate different trade-offs would require the process to be rerun.

The main contribution of this paper is the introduction of an automatic design space exploration approach, based on genetic algorithms, for reliable systems implemented on SRAM-based FPGAs, exploiting classical fault tolerance techniques coupled with partial dynamic reconfiguration. The paper is focused on problem modeling, in order to define a flexible framework that can be easily extended and integrated with other mitigation techniques and metrics. In particular, the proposed methodology models the problem of reliable system design and identifies significant figures of merit for comparing different solutions; such set of parameters can be further personalized based on the designer's experience and needs in the specific situations.

It is worth noting that the particular platform here considered, introduces yet another degree of complexity, since even though these techniques are applied correctly, from a functional point of view, an opportune, ad-hoc placement and routing process is necessary to guarantee that the functional detection/correction properties hold after FPGA mapping. The problem we tackle aims at exploring the solution space and estimating costs. During the implementation phase, then, it will be necessary to adopt suitable approaches, such as the RoRA one [9]; the authors have developed a reliability-oriented place and route algorithm for guaranteeing that, in the specific case of the TMR technique, no SEU fault causes multiple modules to fail, thus invalidating the adopted reliability technique.

The paper is organized in the following way. Section 2 lays the motivations and background for the proposed methodology and framework, identifying the requirements that have led to their definition and implementation. Section 3 introduces the framework and its characterizing elements; an in-depth discussion of the adopted metrics is presented in Section 4. The proposed approach has been applied to some test circuits to evaluate its effectiveness; the results are discussed in Section 5. Concluding remarks and future work are summarized in the last section.

2 Motivations and Background

The problem of partitioning the system into blocks of logic for applying the TMR fault mitigation technique has been previously investigated in [3]. The authors observe that by using different partitioned groups of logic, here called clusters, as the basis for triplicating and voting, different levels of overheads can be achieved, in terms of area, performance and robustness. In the presented case study, the authors select three different schemas, namely "maximum logic partition", "medium logic partition" and "minimum logic partition", and compare the achieved solutions with respect to the identified parameters.

In our own experience, given a circuit such as Noekeon [10], the trivial solutions corresponding to (a) the triplication of the entire system, which we will label "minimum logic partition" to be aligned with the previous approach, or to (b) the triplication of each

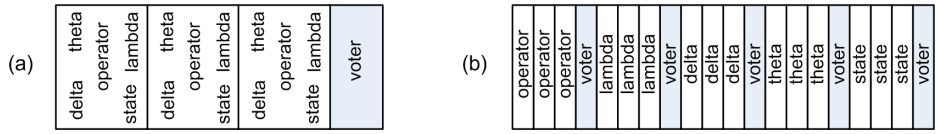

Figure 1. (a) minimum logic partition, and (b) maximum logic partition.

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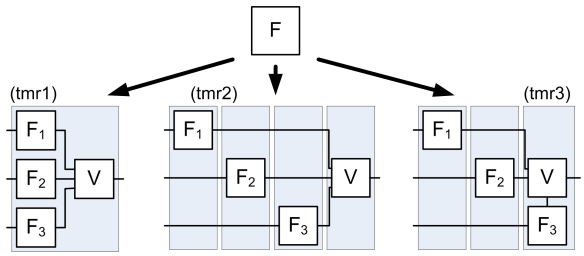

Figure 2. Application of the TMR with different types of mapping.

component in the system, labeled "maximum logic partition", do not identify the most interesting alternatives the designer has. More precisely, there are 52 possible partitionings of the five components, which to apply the TMR to, leading to different area and performance characterization, especially due to the impact of voters, whose dimension depends on the number of lines they monitor; in particular, the most interesting ones are a few solutions corresponding to clusters constituted by two components, as later discussed in Section 5. An additional aspect that comes into play in the present context is the fact that the different replicas can be mapped onto different independently reconfigurable areas in different ways (Figure 2), used by the partial dynamic reconfiguration. In particular, voters not only perform the majority vote, but also identify the faulty replica, so that it is possible to trigger a specific re-writing of the corrupted configuration bits, where necessary. The number of possible solution thus farther grows, reaching the total number of 1866 alternatives (in the specific example) to be compared and evaluated. As a result, additional aspects need be taken into account with respect to [3], and the issue of identifying the most convenient solution with respect to the interesting parameters becomes even more complex.

In this paper, the framework applies the 1-Dimensional (1D) reconfiguration possibility, supported for Xilinx's Virtex2PRO Family [11, 7], yet the problem has been modeled and the framework implemented to support also 2D reconfiguration for more recent platforms and associated design flows.

The overall scenario thus sees the need for a way to explore the solution space by evaluating different partitioning solutions, aiming at identifying the optimal application of reliable techniques coupled with partial reconfiguration on the system specification in order to obtain a hardened FPGA-based system, comparing possible alternatives from different points of view. The input information consists of the circuit to be implemented, the available set of reliability-oriented techniques and the parameters the designer wants to analyze; the expected output is the (set of) solution that best fits the given cost function. The next two sections present these aspects in detail.

3 Methodology Overview

Figure 3 presents an overview of the proposed framework; it consists of three main steps performing 1) an analysis and characterization of the nominal system specification in input, 2) the design space exploration for obtaining a reliable version of the system, and 3) the implementation and the synthesis of the identified reliable system. By following the core based design approach commonly adopted for FPGA-based system design, the circuit is specified through a structural description composed by functional units interconnected by signals; the adopted language is VHDL, restricted to the subset for structural specifi-

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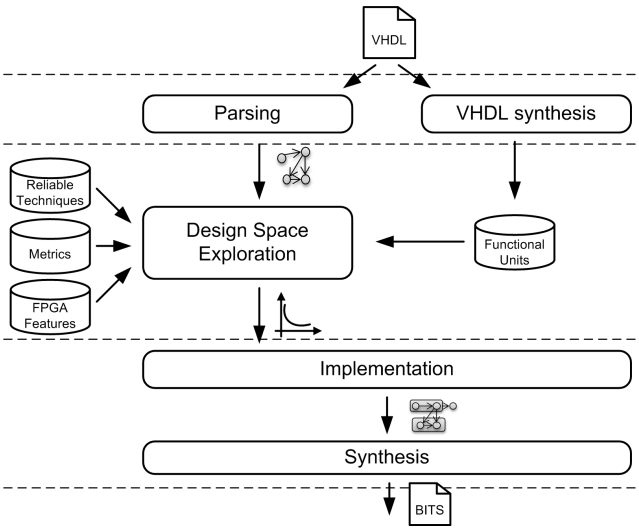

Figure 3. The proposed reconfiguration for reliability framework.

cation. Moreover, the description is organized hierarchically, where each functional unit can be modeled with another structural description or is considered as a black box, if the functionality is described with a process, and wrapped in a new component.

In the first step, the VHDL description is parsed to build an agile internal representation based on graphs where nodes represent functional units and edges interconnections. At the same time, an automatized synthesis process of the VHDL is performed (by using Xilinx's tool-chain) for determining the implementation costs and the characteristics of each type of functional unit composing the structural description; all information is stored in a *functional units' repository.*

The second step represents the core of the proposed methodology and aims at performing the design space exploration for identifying the most promising hardened versions of the considered circuit. Together with the internal representation of the considered circuit, this step takes in input several repositories containing information necessary for performing the design space exploration; they are described in the following paragraphs.

Functional Units' Repository. It contains the descriptions of all the classes of functional units of the current design, together with those of standard units (the most relevant are TSC components such as voters and checkers) and of the FPGA components (such as tri-state buffers, I/O buffers and hardwired multipliers). Each description consists of the specification of the interface (ports) and the list of required FPGA resources (e.g., the number of slices, BRAMs and hardwired components).

Device features' repository. It stores a description of all the various characteristics and features for each considered FPGA device, including the available resources; in particular, the number of available basic reconfigurable cells, in terms of slices, is considered together with the height and width of the grid of basic cells composing the device. Moreover, the constraint on the reconfigurable areas' shape is stored: when considering the module-based 1D reconfigurability approach, each reconfigurable area has a rectangular shape with a height equal to the device grid and a variable width multiple of a fixed number of columns (usually 4) [7]; therefore, the repository entry contains the basic area size, and an indicative maximum number of reconfigurable areas.

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Techniques' Repository. It contains a model of each reliable technique provided in the framework for hardening the system. The techniques are modeled in terms of the required modification on the circuit specification and binding on FPGA resources. In this work, three different implementation schemata of the TMR technique have been considered: as shown in Figure 2, the three techniques modify the specification by triplicating the part of the circuit and by introducing a voter of the correct size; they differ from each other by the imposed mapping constraints. In fact, the first version maps the three replicas and the voter onto the same reconfigurable area, the second one maps the four modules on different reconfigurable areas while the last one clusters the voter with one of the three replicas.

Metrics' Repository It contains a set of metrics the exploration phase uses for estimating in advance characteristics and costs of the hardened versions of the circuit.

The solution space is composed by all reliable implementations of the input circuit, obtained by applying the TMR (and in the future a wider set of techniques) at different granularity levels on the nominal specification. As shown in Figure 4(a), the strategy adopted for generating a reliable solution consists of two steps: first, the functional units of the circuit are partitioned into several clusters; then, on each cluster, a specific technique selected from the repository is applied. It is worth noting that not all possible combinations of clusters and techniques are feasible: in fact, it may happen that for a solution some constraints, such as the ones defined by the FPGA characterization (device dimension or number of reconfigurable areas) are not satisfied, and thus, this alternative is discarded.

A multi-objective strategy has been adopted for performing the design space exploration; the aim is to define a versatile approach to lead the exploration in several directions for finding different solutions, all interesting but with different properties. For the same reason, objective functions (usually one or two) are not predefined but can be specified as the minimization (or maximization) of a single metric or of a combination of metrics available in the repository, thus giving the designer the opportunity to specify the figures of merit to be used for expressing and comparing costs and benefits, on the basis of his/her necessities. Because of the multi-objective approach, the method adopted for comparing different solutions is the Pareto dominance: when minimizing with respect to all figures of merit $\{1, ..., n\}$, a solution vector g is better than solution vector h if and only if $\forall i \in \{1, ...n\} : g_i \leq h_i \wedge \exists j : g_j < h_j$. It is worth noting that the Pareto dominance is defined as a partial order relationship; therefore, when considering two solutions k and j, i) k may dominate j or ii) j may dominate k or iii) k and j may be incomparable (or indifferent).

A multi-objective genetic algorithm, called SPEA2 [12], has been adopted for implementing the design exploration engine. The choice of a genetic algorithm is motivated by the fact

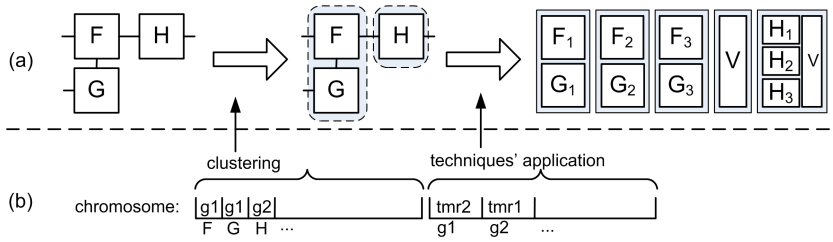

Figure 4. Strategy for generating a reliable solution: (a) the approach, and (b) the solution representation.

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that this kind of evolutionary strategies is widely used for search and optimization problems due to its performance and efficiency. When considering the presented approach for generating a reliable version of the system, the solution is coded in a chromosome composed by two different parts (shown in Figure 4(b)). The first part, characterized by a number of positions equal to the number of functional units in the circuit, is used for specifying the cluster which each functional unit, considered as a black box, is assigned to. Each position of the second part of the chromosome represents a cluster of components and its value is the ID of the technique applied to it. The size of the second part of the chromosome is set as the maximum number of clusters for which there is a feasible solution. Evolution operators do not present particular aspects in their implementation:

The **selection operator** is defined by the SPEA2 algorithm with an approach based on the Pareto dominance;

The **mutation operator** has been implemented to flip the value contained in a random position of the chromosome, implying either a change in the assignment of the cluster of a functional unit or in the type of technique applied to a cluster;

The **crossover operator** has been implemented as a single point crossover which cuts in a specific point the chromosomes of the two parents and recombines subparts for generating new children.

The standard workflow of the genetic algorithm is composed by the generation of an initial population of feasible solutions and the iteration of the execution of the three operators on the population for a given number of generations; unfeasible solutions produced during the evolution are tagged with an infinite cost and are automatically discarded by the algorithm during subsequent generations. Finally, at the end of the execution, the set of solutions – computed on the final population – which are not dominated by any other one (the *Pareto front*) represents the set of best solutions achieved by the algorithm with respect to the specified figures of merit.

During the last step of the framework flow, the designer manually analyzes the structure and the characteristics of the best solutions, computed by the algorithm, by means of all available metrics and chooses the preferred one to be implemented. The VHDL description of the final solution is then automatically generated and synthesized by following the reconfigurable system design flow [13, 14]. In this phase, a reliability oriented place and route strategy [9] should be taken into account.

A first framework prototype has been implemented for supporting the automation of the design space exploration; it is available for download at [13]. Current efforts are devoted to integrating the proposed framework with reconfigurable system design flow proposed by the Earendil methodology [13, 14].

4 Design Metrics

We have identified a set of metrics that are usually taken into account when comparing different solutions; in particular, we have selected and refined the ones adopted in our previous works and in the Earendil methodology; nevertheless, should the designer wish to add new ones, the proposed framework allows it, through the *metrics' repository*.

R_{fu_set}: it represents the number of resources required for implementing a set of functional units mapped onto the same reconfigurable area; it is computed as the sum of imple-

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mentation costs obtained by the initial synthesis of the functional units. Currently, it is represented by the number of slices but it can be extended by considering other FPGA resource types.
D_{rec_area}: it represents the dimension of a reconfigurable area, estimated as the minimum rectangular area satisfying the reconfigurable systems' implementation constraints [7] and containing resources for implementing the set of functional units mapped onto.
R_{system}: it represents the sum of resources required for implementing the entire system.
D_{system}: it represents the dimension of the system, obtained as the sum of D_{rec_area} for all reconfigurable areas belonging to the system.
N_{areas}: it represents the number of reconfigurable areas contained into the system. It is constrained by an upper bound specified in the FPGA characterization.
ro-index: it is a fragmentation index, previously introduced in [6], aiming at estimating the degree of actual utilization of a reconfigurable area' resources. It is computed as ration between R_{fu_set} and D_{rec_area} of a specific area.

These parameters evaluate the solutions in terms of the global impact of the techniques' application on the system and from a local point of view by estimating the characteristics of each reconfigurable area of the system. The designer can use them for performing an overall evaluation of a solution, or can choose a subset of them, for defining the objective functions for the design space exploration.

5 Experimental Session

The proposed framework has been tested on two circuits, a Noekeon cipher and a FM Digital Signal Demodulator, running two sessions using different figures of merit. The aim of the first session (labeled with a) is to achieve a final system characterized by small and balanced areas, quickly reconfigurable in case of a fault, with limited internal fragmentation; the adopted objective functions maximize the average ro-index and the minimize a linear combination of the normalized average area dimension and its standard deviation. In the second session, the goal is to achieve a small system (minimization of the system dimension), still with small and balanced areas (minimization of the linear combination used in the first session). Figures 5 and 6 show the experimental results; marked points represent the optimal solutions.

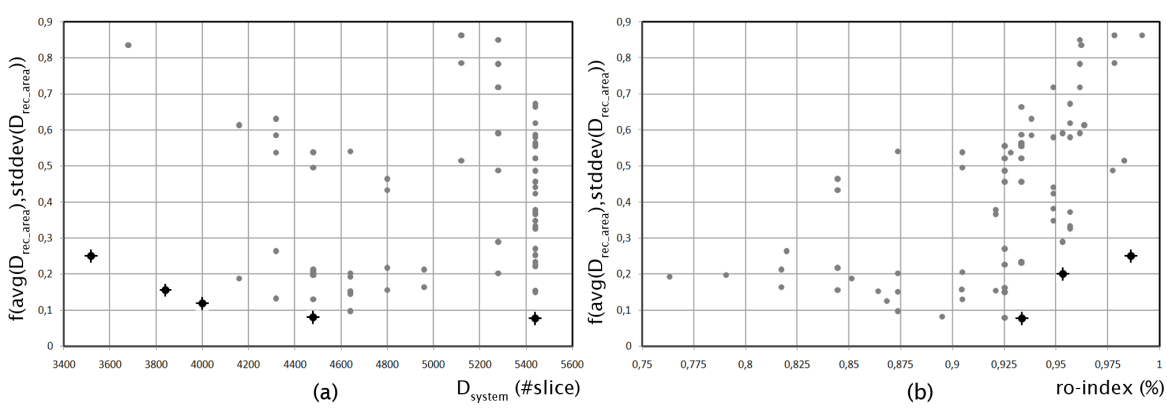

Figure 5. Experimental results on the Noekeon cipher circuit

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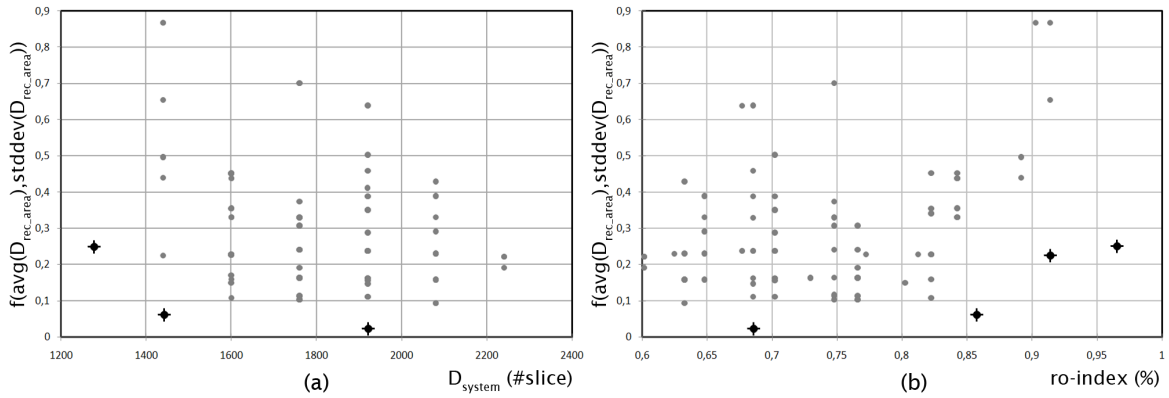

Figure 6. Experimental results on the FM Digital Signal Demodulator circuit

For the example introduced in Section 2, in session (a) on the Noekeon circuit one of the most promising solutions corresponds to the following partitioning: Cluster 0 includes only component `theta`; Cluster 1 includes components `operator`, `delta` and `state`; and finally Cluster 2 includes component `lambda`. The TMR is applied with schema *tmr1* (Figure 2) to Cluster 0 and also to Cluster 2, while *tmr2* is applied to Cluster 1. There are 6 independently reconfigurable areas, having a dimension between 800 and 1120 slices, for a total cost of 4862 slices (and a system dimension of 5440). The trivial solution of triplicating the entire system, mapping each replica on different areas (*tmr2*) for a combination of figures of merit falls into the set of optimal solutions, characterized by a low total area, but having bigger reconfigurable areas, such that recovery from a fault will require a wider action.

From these experiments, as well as from others here not reported, it is possible to draw some conclusions on the proposed framework.

Number of possible solutions. The application of the TMR at different levels of granularity produces a relevant number of possible solutions, differing with respect to the considered parameters; thus an automatic framework such as the proposed one, is necessary to perform the solution space exploration.

Different figures of merit identify different optimal solutions. The identification of different parameters to characterize the possible solutions is a difficult task, considering that there are some dependencies; the selected set of parameters lead to different optimal solutions thus giving the designer an instrument to drive the exploration space.

Optimal solution filtering. The designer can further reduce the number of solutions of the Pareto front (totaling at most 6 in our examples) by filtering on desired cut-off values of the objective functions; for instance, if the designer is interested in solutions with a balanced trade-off among the several objective function, s/he can delete the extreme points of the Pareto front.

6 Conclusions

The paper presented a design space exploration approach for the implementation of SRAM-based FPGAs able to mitigate the effects of SEU faults, both in their configuration and application memory cells, achieved by applying standard fault tolerance techniques and exploiting the partial, dynamic reconfiguration. We have identified a set of figures of

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metric that can be used to compare the various possible solutions, derived by applying the Triple Modular Redundancy at different levels of granularity, i. e., to the entire system or to every module composing the system, or to clusters of modules.

Results show that the number of available solution is quite relevant, and the solution is rarely a trivial one, thus motivating the use of such automatic tool. Among the several identified solution(s) according to the designer's selected important parameters, the framework produces the modified VHDL of the selected solution including the necessary redundancy and the additional information required for correctly mapping the system onto the FPGA, distributing it in the identified reconfigurable areas.

Present work is devoted to extend the set of available reliability oriented design techniques, and to allow the designer to the express his/her own set of figures of merit and design constraints, to explore the solution space.

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